Electronic Engineering Project Specification

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Electronic Engineering Project Specification (A04231)

Short Title: Electronic Eng Proj Spec
Faculty Science and Computing
Department Engineering Technology
Cluster Electronics
Author CLAIRE.FITZPATRICK
Credits 5 **Level:** Advanced

Description of Module / Aims

The aim of the final year project subject is to augment the theoretical material, covered in the course, by exposing the student to circuit design and software development in a work laboratory environment. The benefits derived from co-operating with industry are recognised and where possible industrial based projects will be considered for assigning as student projects. In this semester, the students are encouraged to follow safe systems and working practices and to apply appropriate management techniques (personal and institute resource management, deciding achievable goals and sticking to them, etc.) and to develop their knowledge of the chosen topic, by performing literature and Internet searches and interviewing appropriate staff.

Programmes

`PROJ-0094` BEng (Hons) in Electronic Engineering (`WD_EONIC_B`) 4 / 7 / M

Indicative Content

- Projects are selected in collaboration with industry, based on research themes carried out by researchers at postgraduate level, in collaboration with other academic institutions and on occasion at the suggestion of the student themselves.
- Literature Research (Periodicals, IEEE search engines, Internet)
- Project Management (appraise the problem, break into constituent parts, identify milestones, based on time, difficulty, other resources available and overall ability to achieve the goals)
- Early Milestones
- Presentation through seminar / report on progress

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Perform a thorough literature search including on-line scientific engineering databases, in a given area and compile the salient expressions, synopses, alternative techniques and their own proposed approach.
2. Write realistic goals, based on available resources, difficulty of tasks and required learning/existing abilities for the milestones required to finally complete a project.
3. Defend a proposed course of action, based on rational reasoning, resources and abilities.
4. Present a project context, status quo and projected outcome to peers and staff in an open forum/seminar.
5. Organise component orders, within the project budget and complying with the Institute's process for placing orders and recognising suppliers.
6. Develop and maintain a professional relationship with the supervisor and maintain a log of meetings.

Learning and Teaching Methods

- One-to-one weekly scheduled meetings with supervisor
- Problem-based learning (posing of project-specific problems where student researches and devises an appropriate solution, typically over a 1- or 2-week period)
- Independent learning through a thorough reading of the project area.

Assessment Methods

	Weighing	Outcomes Assessed
Final Project	100%	1,2,3,4,5,6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	48	
Independent Learning	87	